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CST476 MOBILE COMPUTING - Module IV

CST476 MOBILE COMPUTING

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Mobile network layer

The protocols and mechanisms developed for the network layer to support mobility. It provides protocol enhancement that allows transparent routing of IP datagrams to mobile nodes in the internet. Mobile IP adds mobility support to the internet network layer protocol IP. RFC 2002 is a reference document for the complete detail about the mobile IP.

Mobile Internet Protocol (IP)

Mobile IP is a communication protocol (created by extending Internet Protocol, IP) that allows the users to move from one network to another with the same IP address. It ensures that the communication will continue without the user's sessions or connections being dropped. Mobile IP is a communication protocol (created by extending Internet Protocol, IP) that allows the users to move from one network to another with the same IP address.

☐ Requirements for Mobile IP

1. [Transparency](#)

- Mobility should remain 'invisible' for many higher layer protocols and applications. For TCP, mobile computer must keep its IP address.

2. [Compatibility](#)

- Support of the same layer 2 protocols as IP.
- No changes to current end-systems and routers required.
- Mobile end-systems can communicate with fixed systems.

3. [Security](#)

- The minimum requirement: all the messages related to the management of Mobile IP are authenticated.

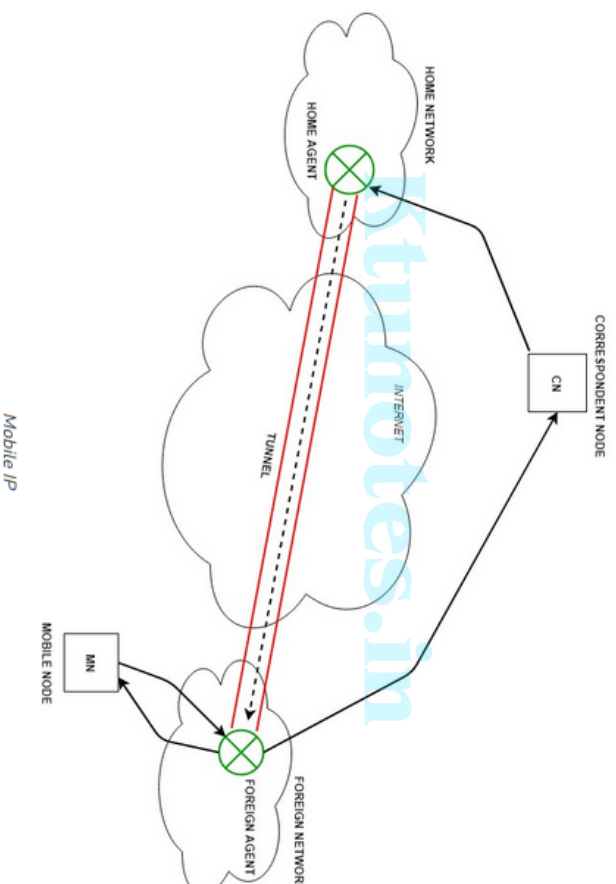
4. [Efficiency and scalability](#).

- Only little additional messages to the mobile system required (connection typically via a low bandwidth radio link).

Mobile IP Design Goals

1. A mobile node must be able to communicate with other nodes after changing it's link-layer attachment, yet without changing its IP address.
2. A mobile node must be able to communicate with other nodes that do not implement mobile IP.
3. Mobile IP must use authentication to offer security against redirection attacks.

4. The number of administrative messages should be small to save bandwidth & power.
5. Mobile IP must impose no additional constraints on the assignment of IP addresses.



Terminologies:

1. Mobile Node (MN) is the hand-held communication device that the user carries e.g. Cell phone.
2. Home Network is a network to which the mobile node originally belongs as per its assigned IP address (home address).
3. Home Agent (HA) is a router in-home network to which the mobile node was originally connected
4. Home Address is the permanent IP address assigned to the mobile node (within its home network).
5. Foreign Network is the current network to which the mobile node is visiting (away from its home network).
6. Foreign Agent (FA) is a router in a foreign network to which the mobile node is currently connected. The packets from the home agent are sent to the foreign agent which delivers them to the mobile node.
7. Correspondent Node (CN) is a device on the internet communicating to the mobile node.
8. Care-of Address (COA) is the temporary address used by a mobile node while it is moving away from its home network.

9. Foreign agent COA, the COA could be located at the FA, i.e., the COA is an IP address of the FA. The FA is the tunnel end-point and forwards packets to the MN. Many MN using the FA can share this COA as a common COA.
10. Co-located COA, the COA is co-located if the MN temporarily acquired an additional IP address which acts as COA. This address is now topologically correct, and the tunnel endpoint is at the MN. Co-located addresses can be acquired using services such as DHCP.

Working

The correspondent node sends the data to the mobile node. Data packets contain the correspondent node's address (Source) and home address (Destination). Packets reach the home agent. But now mobile node is not in the home network, it has moved into the foreign network. The foreign agent sends the care-of-address to the home agent to which all the packets should be sent. Now, a tunnel will be established between the home agent and the foreign agent by the process of tunneling.

Tunneling establishes a virtual pipe for the packets available between a tunnel entry and an endpoint. It is the process of sending a packet via a tunnel and it is achieved by a mechanism called encapsulation.

Now, the home agent encapsulates the data packets into new packets in which the source address is the home address and destination is the care-of-address and sends it through the tunnel to the foreign agent. Foreign agent, on another side of the tunnel, receives the data packets, decapsulates them, and sends them to the mobile node. The mobile node in response to the data packets received sends a reply in response to the foreign agent. The foreign agent directly sends the reply to the correspondent node.

Key Mechanisms in Mobile IP

- **Agent Discovery:** Agents advertise their presence by periodically broadcasting their agent advertisement messages. The mobile node receiving the agent advertisement messages observes whether the message is from its own home agent and determines whether it is in the home network or foreign network.
- **Agent Registration:** Mobile node after discovering the foreign agent sends a registration request (RREQ) to the foreign agent. The foreign agent, in turn, sends the registration request to the home agent with the care-of-address. The home agent sends a registration reply (RREP) to the foreign agent. Then it forwards the registration reply to the mobile node and completes the process of registration.
- **Tunneling:** It establishes a virtual pipe for the packets available between a tunnel entry and an

endpoint. It is the process of sending a packet via a tunnel and it is achieved by a mechanism called encapsulation. It takes place to forward an IP datagram from the home agent to the care-of-address. Whenever the home agent receives a packet from the correspondent node, it encapsulates the packet with source address as home address and destination as care-of-address.

Dynamic Host Configuration Protocol (DHCP)

Dynamic Host Configuration Protocol (DHCP) is a network management protocol used to dynamically assign an IP address to any device, or node, on a network so they can communicate using IP (Internet Protocol). DHCP automates and centrally manages these configurations.

DHCP stands for Dynamic Host Configuration Protocol. It is the critical feature on which the users of an enterprise network communicate. DHCP helps enterprises to smoothly manage the allocation of IP addresses to the end-user clients' devices such as desktops, laptops, cellphones, etc. is an application layer protocol that is used to provide:

Why DHCP?: *DHCP helps in managing the entire process automatically and centrally. DHCP helps in maintaining a unique IP Address for a host using the server. DHCP servers maintain information on TCP/IP configuration and provide configuration of address to DHCP-enabled clients in the form of a lease offer.*

How DHCP works?

DHCP runs at the application layer of the TCP/IP protocol stack to dynamically assign IP addresses to DHCP clients/nodes and to allocate TCP/IP configuration information to the DHCP clients. Information includes subnet mask information, default gateway, IP addresses and domain name system addresses.

DHCP is based on client-server protocol in which servers manage a pool of unique IP addresses, as well as information about client configuration parameters, and assign addresses out of those address pools.

The DHCP lease process works as follows:

- First of all, a client (network device) must be connected to the internet.
- DHCP clients request an IP address. Typically, client broadcasts a query for this information.
- DHCP server responds to the client request by providing IP server address and other configuration information. This configuration information also includes time period, called a lease, for which the allocation is valid.
- When refreshing an assignment, a DHCP clients request the same parameters, but the DHCP server may assign a new IP address. This is based on the policies set by the administrator.

Benefits of DHCP

There are following benefits of DHCP:

Centralized administration of IP configuration: DHCP IP configuration information can be stored in a single location and enables that administrator to centrally manage all IP address configuration information.

Dynamic host configuration: DHCP automates the host configuration process and eliminates the need to manually configure individual host. When TCP/IP (Transmission control protocol/Internet protocol) is first deployed or when IP infrastructure changes are required.

Seamless IP host configuration: The use of DHCP ensures that DHCP clients get accurate and timely IP configuration IP configuration parameter such as IP address, subnet mask, default gateway, IP address of DNS server and so on without user intervention.

Flexibility and scalability: Using DHCP gives the administrator increased flexibility, allowing the administrator to move easily change IP configuration when the infrastructure changes.

Mobile Adhoc Network (MANET)

MANET stands for Mobile Adhoc Network also called a wireless Adhoc network or Adhoc wireless network that usually has a routable networking environment on top of a Link Layer ad hoc network. They consist of a set of mobile nodes connected wirelessly in a self-configured, self-healing network without having a fixed infrastructure. MANET nodes are free to move randomly as the network topology changes frequently. Each node behaves as a router as they forward traffic to other specified nodes in the network.

MANET may operate a standalone fashion or they can be part of larger internet. They form a highly dynamic autonomous topology with the presence of one or multiple different transceivers between nodes. The main challenge for the MANET is to equip each device to continuously maintain the information required to properly route traffic.

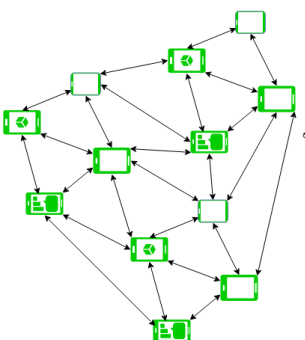


Figure - Mobile Ad Hoc Network

Characteristics of MANET

Some characteristics of adhoc network are as follows:

- **Dynamic topologies:** nodes are free to move arbitrarily; thus the network topology may be changed randomly and unpredictably and primarily consists of bidirectional links. In some cases where the transmission power of two nodes is different, a unidirectional link may exist.
- **Bandwidth-constrained and variable capacity links:** wireless links continue to have significantly lower capacity than infrastructure networks.
- **Energy-constrained operation:** some or all of the MSs in a MANET may rely on batteries or other exhaustible means for their energy. For these nodes or devices, the most important system design optimization criteria may be energy conservation.
- **Limited physical security:** MANETs are generally more prone to physical security threats than wireline networks. The increased possibility of eavesdropping, spoofing, and denial of services (DoS) attacks should be considered carefully. To reduce security threats, many existing link security techniques are often applied within wireless networks.

Applications of MANET

Some specific applications of ad hoc networks include industrial and commercial applications involving cooperative mobile data exchange. There are many existing and future military networking requirements for robust, IP-compliant data services within mobile wireless communication networks, with many of these networks consist of highly dynamic autonomous topology segments. Advanced features of Mobile ad hoc networks combined with other network structures are enabling new applications.

- **Defense applications:** Many defense applications require on the fly communications set-up, and ad hoc/sensor networks are excellent candidates for use in battlefield management.

- **Crisis management applications:** These arise, for example, as a result of natural disasters in which the entire communication infrastructure is in disarray. Restoring communications quickly is essential.

- **Telemedicine:** The paramedic assisting the victim of a traffic accident in a remote location must access medical records (e.g. X-rays) and may need video conference assistance from a surgeon for an emergency intervention. In fact, the paramedic may need to instantaneously relay back to the hospital the victim's X-rays and other diagnostic tests from the site of the accident.

• **Tele-geoprocessing application:** The combination of GPS, GIS (Geographical Information Systems), and high-capacity wireless mobile systems enables a new type of application referred to as tele-geo processing.

• **Virtual Navigation:** A remote database contains the graphical representation of building, streets, and physical characteristics of a large metropolis. They may also “virtually” see the internal layout of buildings, including an emergency rescue plan, or find possible points of interest.

• **Education via the internet:** Educational opportunities available on the internet or remote areas because of the economic infeasibility of providing expensive last-mile wire line internet access in these areas to all subscribers.

• **Vehicular area network:** This a growing and very useful application of adhoc network in providing emergency services and other information. This is equally effective in both urban and rural setup. The basic and exchange necessary data that is beneficial in a given situation.

Dynamic Source Routing (DSR)

The Dynamic Source Routing protocol (DSR) is a simple and efficient routing protocol designed specifically for use in multi-hop wireless ad hoc networks of mobile nodes. DSR allows the network to be completely self-organizing and self-configuring, without the need for any existing network infrastructure or administration. It is a reactive protocol and all aspects of the protocol operate entirely on-demand basis. It works on the concept of source routing. Source routing is a routing technique in which the sender of a packet determines the complete sequence of nodes through which, the packets are forwarded. The protocol is composed of the two main mechanisms of *Route Discovery* and *Route Maintenance*.

The advantage of source routing is : intermediate nodes do not need to maintain up to date routing information in order to route the packets they forward.

DSR requires each node to maintain a route-cache of all known self-to-destination pairs. If a node has a packet to send, it attempts to use this cache to deliver the packet.

Route Discovery

- It finds out the route from a source node to a destination node.

- When a source node wants to send a message to some destination node, then it first searches for its route cache to find whether there is a route to the destination is already exists or not.

- If there is no route to the destination, then the source node will initiate a Route Discovery and send out

Route Request Message which is broadcasted to all the nodes within its transmission range.

- The Route Request Message contains the destination address, the source address, and a unique identification number.

- Each node that receives the Route Request Message checks whether it has a route to the destination or not. If it does not, it adds its address to the route record of the message and then rebroadcasts the Route Request Message on its outgoing nodes.

- When a message reaches the destination node, it will send a *Route Reply Message* towards the source node and this message contains the source route record list which is collected when the Route Request message is forwarded along its way to the destination.

- When the destination sends the Route Reply it uses MAC protocols such as IEEE 802.11 that require a bidirectional link.

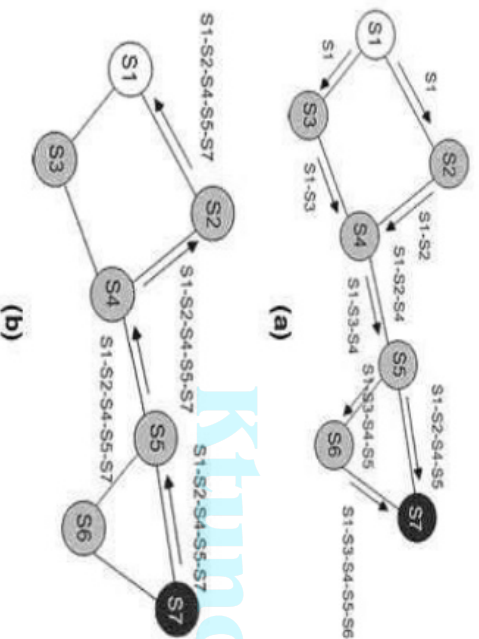
- When the source node receives the Route Reply message, it stores returned route into its route cache. From then on-wards all the messages destined to the same destination will use this route.

Route maintenance

- The ad hoc network is dynamic in nature and the topology of the network changes frequently therefore, existing routes in route cache are broken frequently. Hence, route maintenance is very important.
- After forwarding a message, a node must need to confirm the reachability of the next-hop node.
- If the node does not receive any confirmation from the next hop during a certain period, it will retransmit the packet. If after the number of retransmission still does not receive any confirmation, it will think the link to the next hop is broken and will send a Route Error message to the source node.
- To check this DRS uses three acknowledgment methodologies such as
 - Link-layer acknowledgment - Provided by MAC layer protocol such as IEEE 802.11.
 - Passive acknowledgment - Node hears the next-hop node forwarding the message.
 - Network-layer acknowledgment - Node sends an explicit acknowledgment request to its next-hop.

Example

In the following example, the route discovery procedure is shown where S1 is the source node and S7 is the destination node.



In this example, the destination S7, gets the request through two paths. It chooses one path based on the route records in the incoming packet and sends a reply using the reverse path to the source node. At each hop, the best route with minimum hop is stored. In this example, it is shown the route record status at each hop to reach the destination from the source node. Here, the chosen route is S1-S2-S4-S5-S7.

Advantages of DSR

- Reduces overhead on route maintenance because routes maintain only between nodes involved.
- Route cache also reduces overhead during route discovery time.
- Single route discovery generates numerous routes because intermediate nodes replies their local caches.

Disadvantages of DSR

- Packet header size grows with route length due to source routing.
- A Flood of route requests may reach all nodes in the network.
- Potential collisions between route requests broadcasted by neighboring nodes.
- Insertion of random delays before forwarding RREQ.
- Increased contention because many route replies come back due to nodes replying from their local cache.
- Route Reply may create a Storm problem sometimes.
- Stale caches may cause increased overhead.

Destination Sequence Distance Vector (DSDV)

Destination Sequenced Distance Vector (DSDV) is a hop-by-hop vector routing protocol requiring each node to periodically broadcast routing updates. This is a table driven algorithm based on modifications made to the Bellman-Ford routing mechanism. Each node in the network maintains a routing table that has entries for each of the destinations in the network and the number of hops required to reach each of them. Each entry has a sequence number associated with it that helps in identifying stale entries. This mechanism allows the protocol to avoid the formation of routing loops. Each node periodically sends updates tagged throughout the network with a monotonically increasing even sequence number to advertise its location.

Destination-Sequenced Distance-Vector Routing (DSDV) is a table-driven routing scheme for ad hoc mobile networks based on the Bellman-Ford algorithm. It was developed by C. Perkins and P. Bhagwat in 1994. The main contribution of the algorithm was to solve the routing loop problem.

Every node stores the *next routing hop* for every reachable destination in their routing table. The route used is the one with the highest sequence number i.e. the most recent one. When a neighbor B of A finds out that A is no

longer reachable, it advertises the route to A with an infinite metric and a sequence number one greater than the latest sequence number for the route forcing any nodes with B on the path to A, to reset their routing tables.

Routing table updates in DSDV are distributed by two different types of update packets:

- **Full dump:** This type of update packet contains all the routing information available at a node. As a consequence, it may require several Network Protocol Data Units (NPDUs) to be transferred if the routing table is large. Full dump packets are transmitted infrequently if the node only experiences occasional movement.

- **Incremental:** This type of update packet contains only the information that has changed since the latest full dump was sent out by the node. Hence, incremental packets only consume a fraction of the network resources compared to a full dump.

Selection of Route

If a router receives new information, then it uses the latest sequence number. If the sequence number is the same as the one already in the table, the route with the better metric is used. Stale entries are those entries that have not been updated for a while. Such entries as well as the routes using those nodes as next hops are deleted.

Advantages

- The availability of paths to all destinations in network always shows that less delay is required in the path set up process.
- The method of incremental update with sequence number labels, marks the existing wired network protocols adaptable to Ad-hoc wireless networks. Therefore, all available wired network protocol can be useful to ad hoc wireless networks with less modification.

Disadvantages

- DSDV requires a regular update of its routing tables, which uses up battery power and a small amount of bandwidth even when the network is idle.
- Whenever the topology of the network changes, a new sequence number is necessary before the network re-converges; thus, DSDV is not suitable for highly dynamic or large scale networks.

Example:

Source: <https://minigranth.in/mobile-adhoc-networks-tutorial/dsdv-routing-protocol-manet>

Ad-hoc routing protocols

An ad hoc routing protocol is a convention, or standard, that controls how nodes decide which way to route packets between computing devices in a mobile ad hoc network.

In ad hoc networks, nodes are not familiar with the topology of their networks. Instead, they have to discover it: typically, a new node announces its presence and listens for announcements broadcast by its neighbors. Each node learns about others nearby and how to reach them, and may announce that it too can reach them. The basic rules is that a new node enters into an ad-hoc network, must announce its arrival and presence.

- 1. **Pro-active routing protocols:** These are also known as table-driven routing protocols. Each mobile node maintains a separate routing table which contains the information of the routes to all the possible destination mobile nodes. Destination Sequenced Distance Vector Routing Protocol (DSDV), Global State Routing (GSR)
- 2. **Reactive routing protocols:** These are also known as on-demand routing protocol. In this type of routing, the route is discovered only when it is required/needed. The process of route discovery occurs by flooding the route request packets throughout the mobile network. It consists of two major phases namely, route discovery and route maintenance. Dynamic Source Routing protocol (DSR).

● **3. Hybrid Routing protocol:** It basically combines the advantages of both, reactive and pro-active routing protocols. These protocols are adaptive in nature and adapts according to the zone and position of the source and destination mobile nodes. One of the most popular hybrid routing protocol is Zone Routing Protocol (ZRP).

Characteristics of MANET Routing Protocol:

To avoid the problems with routing in MANET, routing protocols should have following characteristics:

- It should be widely distributed.
- It must be localized.
- Because of nodes mobility, it should be adjustable to frequent change in topology.
- It must be free of impermeable routes.
- The convergence of routes must be fast.
- Each node in the network should be required to store information about the stable local topology.
- It should be able to provide high-quality service.

Mobile Transport Layer

Most application rely on a transport layer such as TCP (transmission control protocol) or UDP (user datagram protocol) in the case of the internet. Not only the lower layers upto network layer support mobility, but also TCP has built-in mechanisms to support mobility.

But UDP does not behave in a network friendly manner, i.e. does not pull back in case of congestion and continues to send packets into an already congested network.

There are many mechanisms in transport layer within TCP that play an important role when using TCP for mobility.

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Traditional TCP- Reliable transport between client and server required. Successful main reason for TCP

Connection oriented, not transaction oriented. Network friendly: time-out congestion and slow down transmission.

Classical TCP Improvements: There are several mechanisms for the classical TCP improvements with the goal to increase TCP's performance in wireless and mobile environments.

- **Indirect TCP:**

Indirect TCP or I-TCP segments the connection and No changes to the TCP protocol for hosts connected to the wired Internet, millions of computers use (variants of) this protocol.

It acts as an optimized TCP protocol for mobile hosts.

The protocol splitting of the TCP connection at, e.g., the foreign agent into 2 TCP connections, no real end-to-end connection any longer All the hosts in the fixed part of the net do not notice the characteristics of the wireless part.

Advantages

- No changes in the fixed network necessary, no changes for the hosts (TCP protocol) necessary, all current optimizations to TCP still work
- Wireless link transmission errors isolated from those in fixed network
- simple to control, mobile TCP is used only for one hop between, e.g., foreign agent and mobile host
- therefore, a very fast re-transmission of packets is possible, the short delay on the mobile hop is known

Disadvantages

- Loss of end-to-end semantics, an acknowledgement to a sender does now not any longer mean that a

receiver really got a packet, foreign agents might crash

- Higher latency possible due to buffering of data within the foreign agent and forwarding to a new foreign agent.

• Snooping TCP:

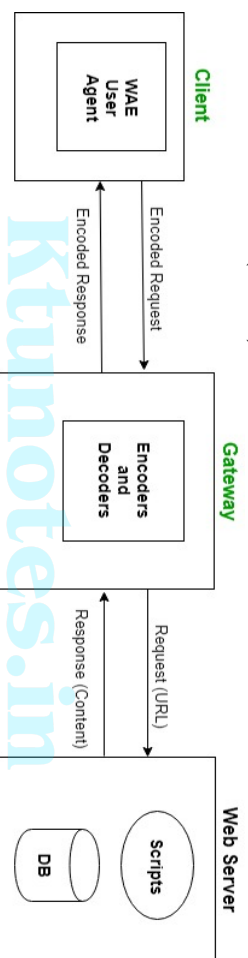
Transparent extension of TCP within the foreign agent where buffering of packets sent to the mobile host and lost packets on the wireless link (both directions!) will be retransmitted immediately by the mobile host or foreign agent, respectively (so called local- retransmission).

The foreign agent therefore snoops the packet flow and recognizes acknowledgements in both directions, it also filters ACKs. The changes of TCP only within the foreign agent.

□ Wireless Application Protocol

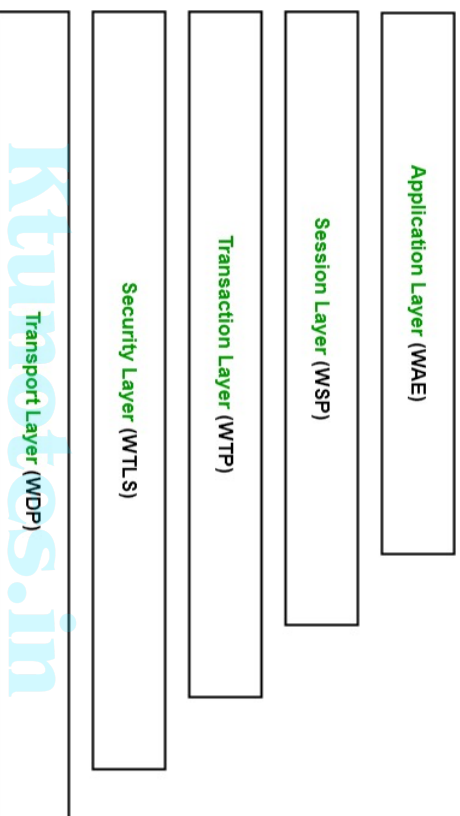
WAP stands for Wireless Application Protocol. It is a protocol designed for micro-browsers and it enables the access of internet in the mobile devices. It uses the mark-up language WML (Wireless Markup Language and not HTML), WML is defined as XML 1.0 application.

In 1998, WAP Forum was founded by Ericson, Motorola, Nokia and Unwired Planet whose aim was to standardize the various wireless technologies via protocols. WAP protocol was resulted by the joint efforts of the various members of WAP Forum. In 2002, WAP forum was merged with various other forums of the industry resulting in the formation of Open Mobile Alliance (OMA).



The user opens the mini-browser in a mobile device. He selects a website that he wants to view. The mobile device sends the URL encoded request via network to a WAP gateway using WAP protocol.

The WAP gateway translates this WAP request into a conventional HTTP URL request and sends it over the internet. The request reaches to a specified Web server and it processes the request just as it would have processed any other request and sends the response back to the mobile device through WAP gateway in WML file which can be seen in the micro-browser.



1. **Application Layer:**

This layer contains the Wireless Application Environment (WAE). It contains mobile device specifications and content development programming languages like WML.

2. **Session Layer:**

This layer contains Wireless Session Protocol (WSP). It provides fast connection suspension and reconnection.

3. **Transaction Layer:**

This layer contains Wireless Transaction Protocol (WTP). It runs on top of UDP (User Datagram Protocol) and is a part of TCP/IP and offers transaction support.

4. **Security Layer:**

This layer contains Wireless Transaction Layer Security (WTLS). It offers data integrity, privacy and authentication.

5. **Transport Layer:**

This layer contains Wireless Datagram Protocol. It presents consistent data format to higher layers of WAP protocol stack.

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☐ **Wireless Datagram Protocol (WDP)**

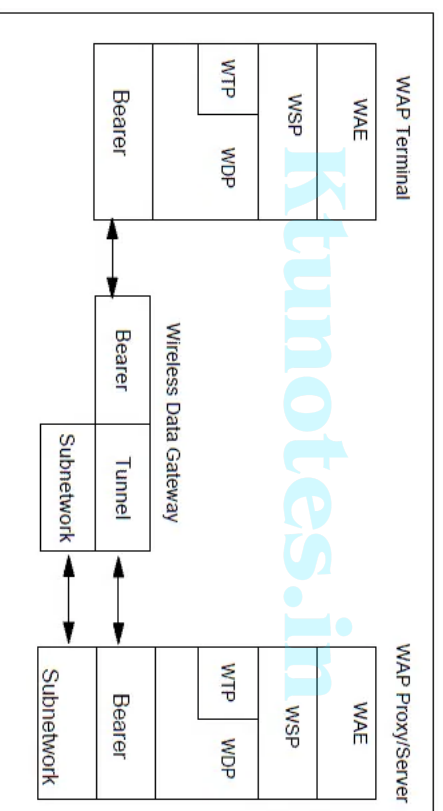
Wireless Datagram Protocol, also known as WDP, provides a consistent service to the upper layers (security, transaction, and session) of the WAP architecture. It is defined in the OMA WAP-259-WDP-20010614-a specification, and allows applications to operate transparently over different available bearer services.

The Wireless Datagram Protocol (WDP) operates on top of many different bearer services capable of carrying data.

At the T-SAP WDP offers a consistent datagram transport service independent of the underlying bearer. To offer

this consistent service, the adaptation needed in the transport layer can differ depending on the services of the bearer.

The closer the bearer service is to IP, the smaller the adaptation can be. If the bearer already offers IP services, UDP is used as WDP. WDP offers more or less the same services as UDP. WDP offers source and destination port numbers used for multiplexing and demultiplexing of data respectively.



The service primitive to send a datagram is TDUinitdata.req with the destination address (DA), destination port (DP), Source address (SA), source port (SP), and user data (UD) as mandatory parameters . Destination and source address are unique addresses for the receiver and sender of the user data.

It communicates transparently over the different bearer services supported by multiple network types. WDP is a connection-less, unreliable datagram service. It supports port number addressing. The port number points to the higher-layer level of WDP. This can be WTP, WTLS, WSP, or an application.

In order to support the different bearer services with its specific capabilities and characteristics, an adaptation is required to keep WDP as a common layer for the various bearer services. Therefore, WDP, with its type of adaptation layer, cooperates with its underlying bearer layer.

WDP messages are sent by the WAP terminal to the wireless data gateway using the bearer services. The wireless data gateway has the choice to pass WDP packets on to the WAP proxy/server through a tunneling protocol, which is the interface between the gateway that provides bearer service and the WAP proxy server.

The WAP proxy/server might offer application content or might operate as gateway between the wireless WTP protocols and the wired Internet.

□ Wireless Transport Layer Security (WTLS)

Wireless Transport Layer Security (WTLS) is a security level for the Wireless Application Protocol (WAP), specifically for the applications that use WAP. It is based on Transport Layer Security (TLS) v1. 0, which is a security layer used in the internet and is a successor to Secure Sockets Layer (SSL) 3.1.

WTLS was developed to address issues surrounding mobile network devices including limited memory capacity, lower processing power and low bandwidth. It also provides authentication, data integrity and privacy protection mechanisms.

Designed to support datagrams in a high-latency, low-bandwidth environment, WTLS provides an optimized handshake through dynamic key refreshing, which allows encryption keys to be regularly updated during a secure session. The method helps clients and servers communicate over a secure and authenticated connection.

The WTLS layer operates above the transport protocol layer. TLS, a standard security protocol used between web browsers and web servers, was modified to develop WTLS. The modification was required because mobile networks could not guarantee end-to-end data security. Consequently, WTLS is optimized for low-bandwidth mobile devices compared to TLS (hence the wireless).

WTLS is more efficient than TLS, and it requires fewer message exchanges. When a message is in the transport layer, WTLS provides privacy management, as well as data authorization and data integrity.

Wireless Transport Layer Security architecture

WTLS includes two layers of protocols.

The WTLS Record Protocol provides basic security to higher layer protocols. It takes user data from the next higher layer and encapsulates it in a protocol data unit.

The following steps are involved in the protocol:

- Initially, a lossless compression algorithm compresses the payload.
- This is followed by the computation and addition of a Message Authentication Code (MAC) over the compressed data using hash-based MAC (HMAC). Hash algorithms like MD5, SHA-1, etc., can be used with the HMAC, with the length of hash code as 0, 5 or 10 bytes.
- The compressed data with MAC is encrypted using a symmetric encryption algorithm like Data Encryption Standard (DES), triple DES, RC5 or the International Data Encryption Algorithm.
- A header gets prepended to the encrypted payload.

The record protocol header consists of record type, record length field indicator, sequence number field indicator, content type and cipher spec indicator.

WTLS also defines several other important higher layer protocols.

□ Wireless Transaction Protocol (WTP)

Wireless transaction protocol (WTP) is a standard used in mobile telephony. It is a layer of the Wireless Application Protocol (WAP) that is intended to bring Internet access to mobile phones. WTP provides functions similar to TCP, except that WTP has reduced amount of information needed for each transaction (e.g. does not include a provision for rearranging out-of-order packets). WTP runs on top of UDP and performs many of the same tasks as TCP but in a way optimized for wireless devices, which saves processing and memory cost as compared to TCP.

It Supports 3 types of transaction:

- Unreliable One-Way Request
- Reliable One-Way Request
- Reliable Two-Way Request

The wireless transaction protocol (WTP) is on top of either WDP or, if security is required, WTLS. WTP has been designed to run on very thin clients, such as mobile phones. WTP offers several advantages to higher layers, including an improved reliability over datagram services, improved efficiency over connection-oriented services, and support for transaction-oriented services such as web browsing.

A transaction is defined as a request with its response, e.g. for a web page. WTP offers many features to the higher layers. The basis is formed from three classes of transaction service as explained in the following paragraphs. Class 0 provides unreliable message transfer without any result message. Classes 1 and 2 provide reliable message transfer, class 1 without, class 2 with, **exactly one reliable result message**.

WTP achieves reliability using duplicate removal, retransmission, acknowledgements and unique transaction identifiers. No WTP-class requires any connection set-up or tear-down phase. This avoids unnecessary overhead on the communication link. WTP allows for asynchronous transactions, abort of transactions, concatenation of messages, and can report success or failure of reliable messages.

To be consistent with the specification, in the following the term initiator is used for a WTP entity initiating a transaction, and the term responder for the WTP entity responding to a transaction. The three service primitives

offered by WTP are TR-Invoke to initiate a new transaction, TR-Result to send back the result of a previously initiated transaction, and TR-Abort to abort an existing transaction.

□ Wireless Session Protocol (WSP)

Wireless Session Protocol (WSP) is an open standard for maintaining high level session. Wireless session that starts when the user connects to one URL and ends when the user leaves that URL.

The wireless session protocol (WSP) has been designed to operate on top of the datagram service WDP or the transaction service WTP. For both types, security can be inserted using the WTLS security layer if required. WSP provides a shared state between a client and a server to optimize content transfer. HTTP, a protocol WSP tries to replace within the wireless domain, is stateless, which already causes many problems in fixed networks.

Wireless Session Protocol (WSP) is an open standard for maintaining high level session. Wireless session that starts when the user connects to one URL and ends when the user leaves that URL. The session wide properties are defined once at the beginning of the session, which has the benefit of saving bandwidth due to the nature of wireless communication. The session establishing process does not have long connection algorithms.

WSP is based on HTTP 1.1 with few enhancements. WSP provides the upper-level application layer of WAP with a consistent interface for two session services. The first is a connection-oriented service that operates above a transaction layer protocol WTP and the second is a connection less service that operates above a secure or non-secure datagram transport service.

- Session management
- Capability negotiation
- Content encoding
- HTTP/1.1 functionality
- Exchange of session headers
- Push and pull data transfer
- Asynchronous requests

WSP/B can run over the transaction service WTP or the datagram service WDP.

END of Module*****